9A 1 REACTION RATE, COLLISION THEORY AND ACTIVATION ENERGY

SPECIFICATION REFERENCE: 9.2 9.3

LEARNING OBJECTIVES

- Know what is meant by the term 'rate of reaction'.
- Calculate the rate of a reaction from:
  (i) the gradient of a suitable graph, by drawing a tangent, either for initial rate or at a time *t*
  (ii) data showing the time taken for a reaction, using
  $\text{rate} = \frac{1}{\text{time}}$.
- Understand that reactions take place only when collisions have sufficient energy, known as the activation energy.

RATE OF REACTION

We can determine the rate of a chemical reaction by the change in concentration of a reactant or a product per unit time.

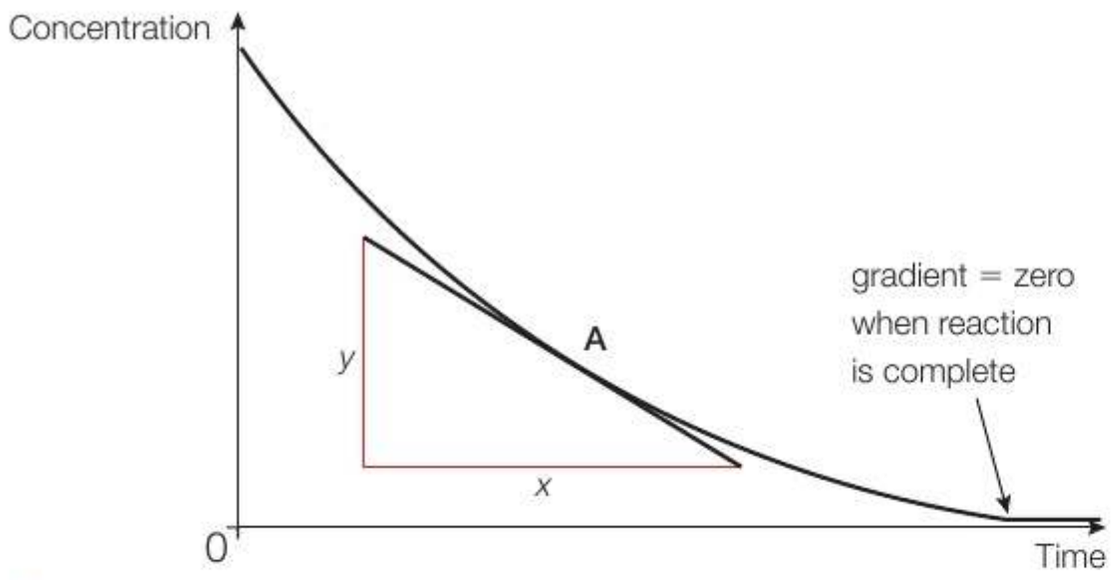
$$\text{rate of reaction} = \frac{\text{change in concentration}}{\text{time for change to happen}}$$

To measure the rate of a reaction, we need to find out:

1 how fast one of the reactants is being used up, or

2 how fast one of the products is being formed.

The graph in **fig A** shows the concentration of a reactant against time, where the gradient (slope) of the graph indicates the rate of the reaction.



▲ **fig A** A graph showing the concentration of a reactant against time.

The gradient decreases as the rate decreases, and becomes zero when all of the reactant is used up.

In the graph in **fig A**, we determine the rate of reaction at point A by drawing a tangent to the curve at point A and measuring its gradient.

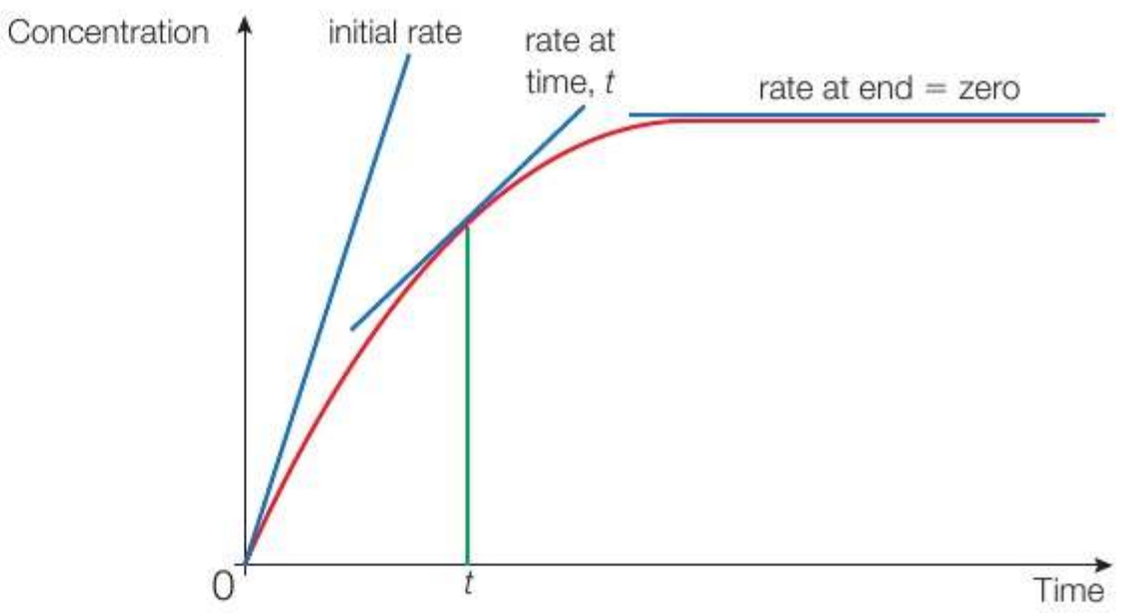
$$\text{gradient} = \frac{y}{x}$$

where y = the change in concentration and x = the change in time.

If the unit of concentration is moles per cubic decimetre (mol dm^{-3}) and the unit of time is seconds (s), then the unit of rate will be moles per cubic decimetre per second ($\text{mol dm}^{-3}\,\text{s}^{-1}$).

Sometimes it is more convenient to measure the concentration of the product formed over a period of time.

The graph in **fig B** shows a plot of concentration against time, with tangents drawn to obtain the initial rate of reaction and the rate at time, *t*.



▲ **fig B** A graph showing the concentration of a product against time.

We can also calculate the rate of reaction from the time taken for a known amount of reactant to be used up, or a known amount of product to be formed. For example, if you measure, in a series of separate experiments, the time taken to collect a given volume of gas, then you can calculate the rate for each experiment using the expression:

$$\text{rate} = \frac{\text{volume of gas collected}}{\text{time taken}}$$

This is illustrated in **table A**.

TIME TAKEN TO COLLECT 20.0 cm^3 OF GAS / s	RATE OF REACTION / $\text{cm}^3\,\text{s}^{-1}$
10.0	2.00
20.0	1.00
40.00	0.50
80.00	0.25

table A The rate of reaction can be calculated from the time taken for a known amount of product to be formed.

COLLISION THEORY

Consider this reaction:

$$A + B \rightarrow C + D$$

In order for molecule A to react with molecule B, the two molecules must first of all collide with each other. If they collide they *may* react.

Why is there a possibility that the molecules may *not* react? This is because not all collisions between reactant molecules will result in a reaction. There are two requirements for a reaction to occur.

- The two molecules must collide with sufficient energy to cause a reaction – the **activation energy E_a**.
- The two molecules must collide in the correct orientation.

ACTIVATION ENERGY, E_a

The activation energy is the minimum energy that colliding particles must possess for a reaction to occur. If the particles collide with less energy than the activation energy, they simply bounce apart and no reaction occurs. Think of the activation energy as a barrier to the reaction. Only those collisions that have energies equal to or greater than the activation energy result in a reaction.

Any chemical reaction results in the breaking of some bonds (needing energy) and the making of new ones (releasing energy). Obviously, some bonds have to be broken before new ones can be made. Activation energy is involved in breaking some of the original bonds.

Where collisions are relatively gentle, there is not enough energy available to start the bond-breaking process, and so the particles do not react.

ORIENTATION

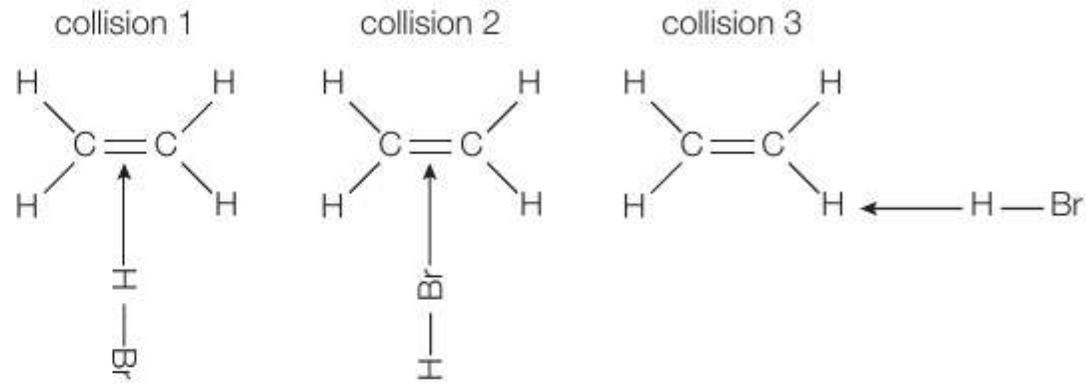
Consider the reaction between ethene and hydrogen bromide, which you met in **Topic 5**:



ethene + hydrogen bromide ⟶ bromoethane

The reaction can only happen if the hydrogen end of the H—Br molecule approaches the C=C of the ethene molecule. Any other collision between the two molecules will result in the molecules simply bouncing off each other.

Of the collisions shown in the figure below, only collision 1 may possibly lead to a reaction.



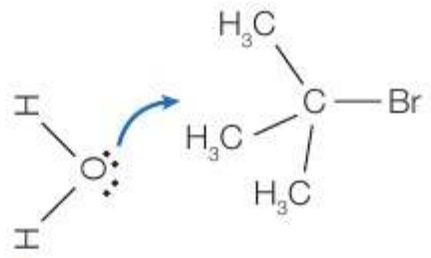
possible reaction no reaction no reaction

DID YOU KNOW?

When the shapes of molecules influence reactions, we say that there is a 'steric factor' involved in the reaction. In some cases, the atoms (or groups of atoms) in a molecule can hinder (interfere with) the course of a reaction. If an atom or group of atoms is particularly large, then it can get in the way of an attacking species. If this happens, we say that the reaction suffers from 'steric hindrance'.

For example, the halogenoalkane 2-bromomethylpropane, $(CH_3)_3CBr$, is hydrolysed rapidly when added to water. However, the mechanism of the reaction does not involve attack by the water molecule on the $\delta+$ carbon atom attached to the bromine, as it does with bromomethane, CH_3Br.

This is because the three methyl groups are so large that they prevent the water molecule approaching the carbon atom sufficiently close to interact with it. The mechanism for the reaction is totally different.



▲ **fig C** The three methyl groups do not allow the water molecule to approach the carbon atom of the C–Br bond.

ADDITIONAL READING

If you would like to find out more, you can research S_N1 and S_N2 mechanisms for the hydrolysis of halogenoalkanes on the internet.

CHECKPOINT

1. At room temperature and pressure, in each cubic decimetre of gas there are about 1×10^{32} collisions every second. Why, therefore, are reactions between gases not completed in a fraction of a second?
2. Why do many reactions involving organic compounds have to be heated or refluxed for long periods of time?
3. Chloroalkanes, such as chloromethane (CH_3Cl), can be hydrolysed to alcohols by heating with aqueous sodium hydroxide. The hydroxide ion acts as a nucleophile attacking the $\delta+$ carbon atom and replacing the chlorine:

   $$CH_3Cl + OH^- \rightarrow CH_3OH + Cl^-$$

   When tetrachloromethane (CCl_4) is heated under reflux with aqueous sodium hydroxide, no reaction takes place. Suggest a reason why.
4. What will happen if a hydrogen atom and a chlorine atom collide and their combined energy of collision is 1.0×10^{-18} J? [E(H—Cl) = 431 kJ mol^{-1}].

SUBJECT VOCABULARY

activation energy, E_a the minimum energy that colliding particles must possess for a reaction to occur
steric hindrance the slowing of a chemical reaction due to large groups within a molecule getting in the way of the attacking species

9A 2 EFFECT OF CONCENTRATION, PRESSURE AND SURFACE AREA ON RATE OF REACTION

SPECIFICATION REFERENCE 9.1 PART

LEARNING OBJECTIVES

- Understand, in terms of the collision theory, the effect of changes in concentration of solution, pressure of a gas and surface area of a solid on the rate of a chemical reaction.

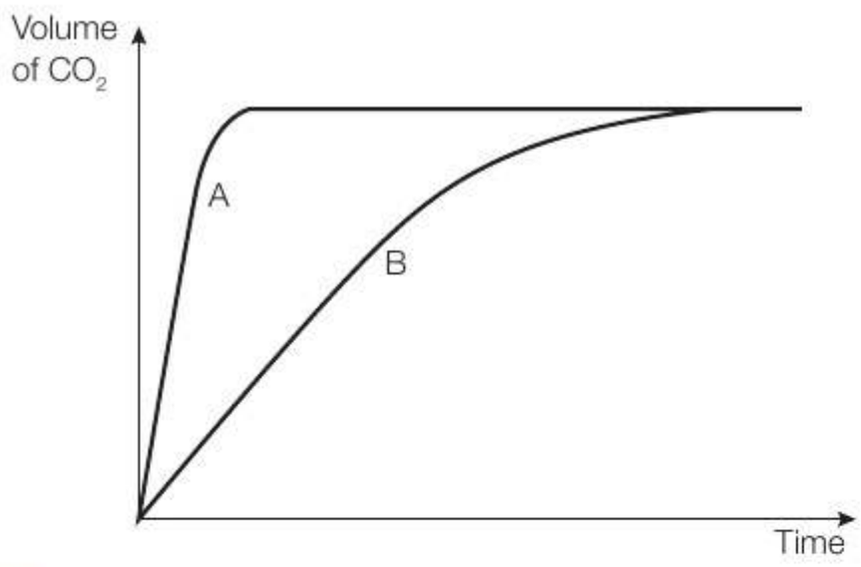
According to the collision theory, reactant particles have to collide with sufficient energy before they can react. It is sensible, therefore, to suggest that we can increase the rate of a reaction by increasing the frequency of collisions with sufficient energy between reactant particles. This is often true, but as in so many situations in chemistry, there are exceptions. You will learn about these in **Topic 11 (Book 2: IAL)**.

Collisions that result in a reaction are called *successful* collisions.

THE EFFECT OF CONCENTRATION

For reactions in solution, an increase in concentration often causes an increase in reaction rate. For many reactions, if the concentration of a solution is increased, then the frequency of collisions between reacting solute particles also increases. This is because they are closer together as there are more of them in a volume of solution. The frequency of successful collisions increases (i.e. there are more successful collisions per second), which in turn produces an increase in the rate of reaction.

The graph in **fig A** shows the effect of the change in volume of carbon dioxide given off with time for the reaction between calcium carbonate and excess dilute hydrochloric acid.



▲ **fig A** A concentration-time graph showing the effect of the change in volume of carbon dioxide given off with time for the reaction between calcium carbonate and excess dilute hydrochloric acid.

Curve A represents the higher concentration of acid. You will notice that the gradient of curve A is always greater than the gradient of curve B. You will also notice that curve A levels off before curve B. Both factors indicate that the rate of reaction is greater for the higher concentration of acid.

THE EFFECT OF PRESSURE

For a reaction in which molecules collide and react in the gas phase, an increase in pressure will cause an increase in the rate of reaction.

The explanation is similar to that for concentration of solution. If the pressure of the gaseous mixture is increased, there will be more reactant molecules in a given volume of mixture. The frequency of collisions will increase. This will result in an increase in the rate of reaction.

Changing the pressure has almost no effect on reactions in the solid or liquid phase. This is because the volume of solids and liquids changes very little when they are put under pressure, so their particles do not move closer together.

ADDITIONAL READING

THE HABER PROCESS

It is interesting to look at the situation that exists when a reaction between two gases takes place on the surface of a solid that is in contact with the reaction mixture. This is the case with the catalytic conversion of hydrogen gas and nitrogen gas to make ammonia in the Haber process.

$$3H_2(g) + N_2(g) \overset{Fe(s)}{\rightleftharpoons} 2NH_3(g)$$

The reaction between the hydrogen and nitrogen molecules takes place at the surface of the iron catalyst, but only at selected sites called 'active' sites. At the high pressures used in the Haber process, there are always more reactant molecules than there are active sites. For this particular reaction, an increase in pressure will not increase the rate of reaction, since all of the active sites are already occupied.

LEARNING TIP

A heterogeneous reaction is one in which the reactants are in more than one phase; for example, a solid and a gas, or a solid and a solution.

THE EFFECT OF SURFACE AREA

For heterogeneous reactions involving a solid, a larger surface area of the solid will result in a faster reaction.

The reaction between magnesium and dilute hydrochloric acid is represented by this ionic equation:

$$Mg(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + H_2(g)$$

Only collisions between the hydrogen ions and magnesium atoms on the *surface* of the magnesium can result in reaction. If the magnesium is powdered, the surface area is increased and hydrogen is given off more quickly.

The effectiveness of solid catalysts is also improved if they are finely divided. For example, the rate of the catalysed decomposition of hydrogen peroxide by manganese(IV) oxide is increased significantly if the catalyst is a powder rather than lumps. Here is the equation for the reaction:

$$H_2O_2(aq) \xrightarrow{MnO_2(s)\ \text{catalyst}} H_2O(l) + \tfrac{1}{2}O_2(g)$$

EXAM HINT

It is important to remember that for a reaction at a position of equilibrium between reactants and products, a catalyst does not affect the position of equilibrium. It only speeds up the rate at which the system achieves equilibrium.

CHECKPOINT

SKILLS REASONING

1. Ammonia and hydrogen chloride react in the gas phase to form ammonium chloride.

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

State and explain the effect, if any, on the rate of reaction of:

(a) halving the volume of the container at constant temperature

(b) increasing the pressure by adding more ammonia at constant volume and temperature.

2. (a) Draw a concentration–time graph for the reaction between marble chips (calcium carbonate) and excess dilute hydrochloric acid. Place the concentration of the hydrochloric acid on the vertical axis.

(b) On the same axes, draw the curve that would be obtained if the reaction was repeated at the same temperature, with the same mass of powdered marble.

(c) Explain the reason for the different-shaped curves obtained.

9A 3 EFFECT OF TEMPERATURE ON RATE OF REACTION

SPECIFICATION REFERENCE 9.4 PART

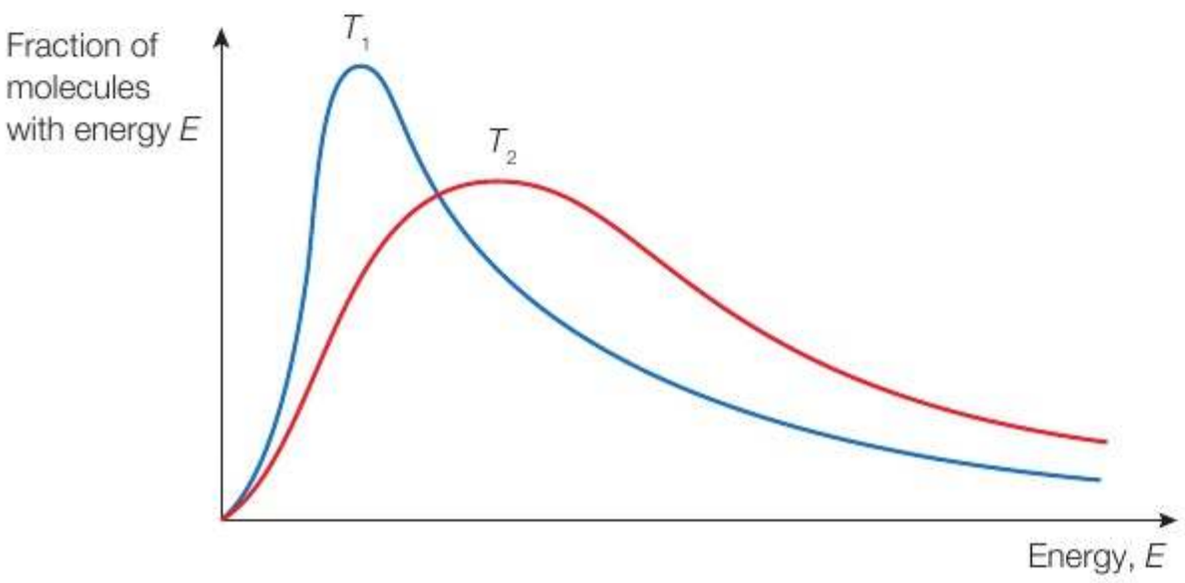
LEARNING OBJECTIVES

- Understand qualitatively, in terms of the Maxwell-Boltzmann distribution of molecular energies, how changes in temperature can affect the rate of a reaction.

MAXWELL–BOLTZMANN DISTRIBUTION CURVES

The molecules in a sample of gas have a wide range of energies. To estimate what fraction of collisions will have the required activation energy, we need to know the energy distribution of the molecules.

This was first calculated in 1860 by James Clerk Maxwell and verified in 1872 by Ludwig Boltzmann. **Fig A** shows the distribution of molecular energies at two temperatures: T_1 (in blue) and T_2 (in red). T_2 is a higher temperature than T_1.



▲ **fig A** A graph showing the distribution of molecular energies at two temperatures. T_2 is a higher temperature than T_1.

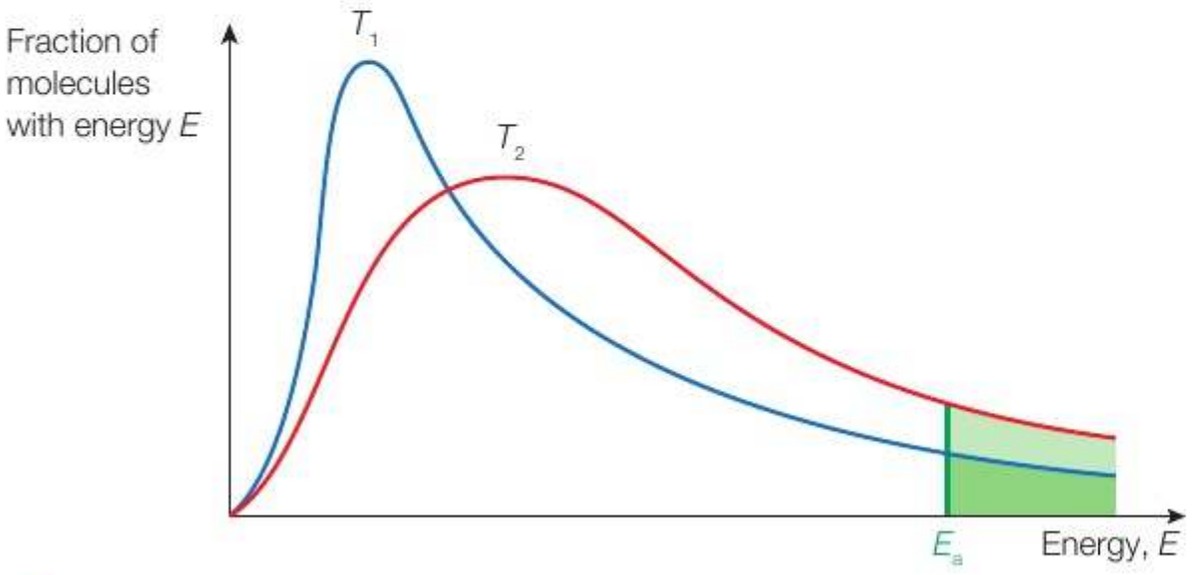
EXAM HINT

Make sure that you label the axes correctly if you are asked to reproduce this graph in an exam.

There are four important points to note about the curves:

- Neither curve is symmetrical.
- Both curves start at the origin and finish by approaching the *x*-axis asymptotically.
- The area under each curve is the same, since the number of molecules has not changed.
- The peak of T_2 is displaced to the right and is lower than the peak of T_1.

The curves in the graph in **fig B**, show, once again, the molecular distribution of energies at two temperatures, T_1 and T_2, where $T_2 > T_1$. E_a is the activation energy for the reaction.



▲ **fig B** A graph showing the distribution of molecular energies at two temperatures, $T_2 > T_1$.

The area shaded dark green represents the fraction of molecules that have the required energy to react at T_1.

The combined area shaded in dark green and light green represents the fraction of molecules that have the required energy to react at T_2.

It is easy to see that the fraction of molecules that can react at the higher temperature is greater.

LEARNING TIP

The Maxwell–Boltzmann diagram showing the effect of temperature consists of *two* curves and *one* activation energy line.

If you are asked to use the Maxwell–Boltzmann distribution curves as part of your explanation, you must make it clear that the area under the curve to the right of the activation energy line represents the fraction of molecules that has enough energy for the collisions to be successful.

HOW CHANGES IN TEMPERATURE CAN AFFECT THE RATE OF A REACTION

We can now explain the effect of an increase in temperature on the rate of a reaction.

An increase in temperature increases the fraction of molecules that possess the required activation energy. The rate of the reaction increases because the number of successful collisions per second increases.

The argument used above assumes that the fraction of collisions with energy greater than or equal to E_a is the same as the fraction of molecules with this energy. This is not exactly true. However, the difference is very small at high energies when the fraction is very small. Under these circumstances, it is reasonable to draw and use the molecular energy distribution curve instead of the collision distribution curve.

LEARNING TIP

When explaining the increase in reaction rate owing to an increase in temperature, it is important to mention that the number of *successful* collisions per second increases, and not just that the *overall* number of collisions per second increases.

Although there is an increase in the overall collision frequency, the effect of this is negligible (very small) compared to the increase in frequency of collisions that have energy equal to or greater than the activation energy.

COLLISIONS IN SOLUTION

In the gas phase, molecules are moving around at high speeds so they frequently collide with one another. At a pressure of 100 kPa and a temperature of 298 K, a single molecule might have somewhere between 10^9 and 10^{10} collisions per second, depending on its size.

In solution, the situation is different. The molecules are much more closely packed together so there is not a lot of 'space' between them. In low-to-medium concentrations, most of the solution is solvent, so solute molecules tend to be entirely surrounded by solvent molecules. These solute molecules are said to be trapped in a 'solvent cage'.

You may think that collisions between solute molecules would be far less frequent than similar collisions in the gas phase because the solvent molecules will get in the way. However, there are situations in which a number of solute molecules become trapped in the same solvent cage. This increases the collision rate between the solute molecules and, if the collisions are sufficiently energetic, they may react, just as in the gas phase.

There are, therefore, two distinct stages to a reaction in solution:

- Firstly, the molecules have to come together, by a process of diffusion, in the same solvent cage.
- Secondly, they have to react.

There may, of course, be some reactions between solute molecules that have 'jumped out' of their cage and just happen to meet each other.

Although what is happening in a solution phase reaction is different from what is happening in the gas phase, the resulting kinetics are the same. For simple reactions involving two species, the chance of them meeting each other in solution is proportional to their concentrations, just as in the gas phase.

CHECKPOINT

1. State the effect of increasing the temperature on the rate of:

(a) an exothermic reaction

(b) an endothermic reaction.

2. (a) Draw Maxwell–Boltzmann distribution curves for a gas at two temperatures, T_c and T_h, where $T_c < T_h$.

(b) Indicate a suitable value for the activation energy of the reaction and use the curves to explain the effect of lowering the temperature on the rate of reaction.

9A 4 EFFECT OF CATALYSTS ON RATE OF REACTION

SPECIFICATION REFERENCE: 9.5 9.6 9.7 9.8

LEARNING OBJECTIVES

- Be able to interpret the action of a catalyst in terms of a qualitative understanding of the Maxwell–Boltzmann distribution of molecular energies.
- Understand the role of catalysts in providing alternative reaction routes of lower activation energy.
- Be able to draw the reaction profiles of both uncatalysed and catalysed reactions including the energy level of the intermediate formed with the catalyst.
- Understand the use of catalysts in industry to make processes more sustainable by using less energy and/or higher atom economy.
- Understand the use of a solid (heterogeneous) catalyst for industrial reactions involving gases, in terms of providing a surface for the reaction.

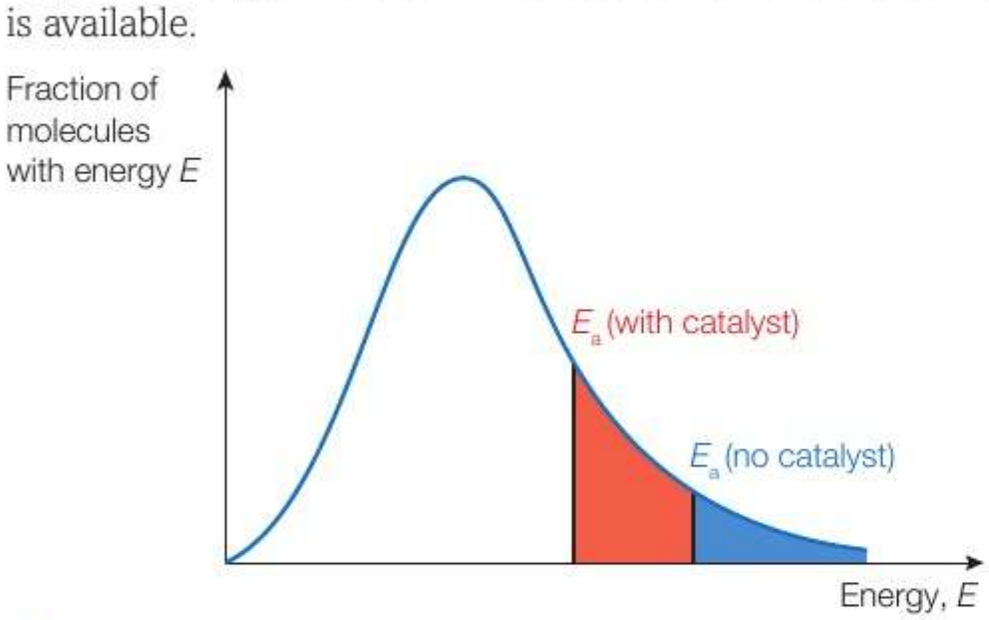
THE EFFECT OF CATALYSTS

A **catalyst** works by providing an alternative route for the reaction. This alternative route has a lower activation energy than the original route.

EXAM HINT

It is wrong to say that a catalyst lowers the activation energy for a particular **reaction pathway**. It provides a *different* pathway with a lower activation energy.

Fig A shows the effect on the fraction of molecules that have the required energy to react when a route of lower activation energy is available.



▲ **fig A** A graph showing the effect on the fraction of molecules that have the required energy to react when a catalyst is present.

The blue shaded area represents the fraction of molecules that have $E > E_a$ when no catalyst is present.

The combined blue and red shaded areas represent the fraction of molecules that have $E > E_a$ when a catalyst is present.

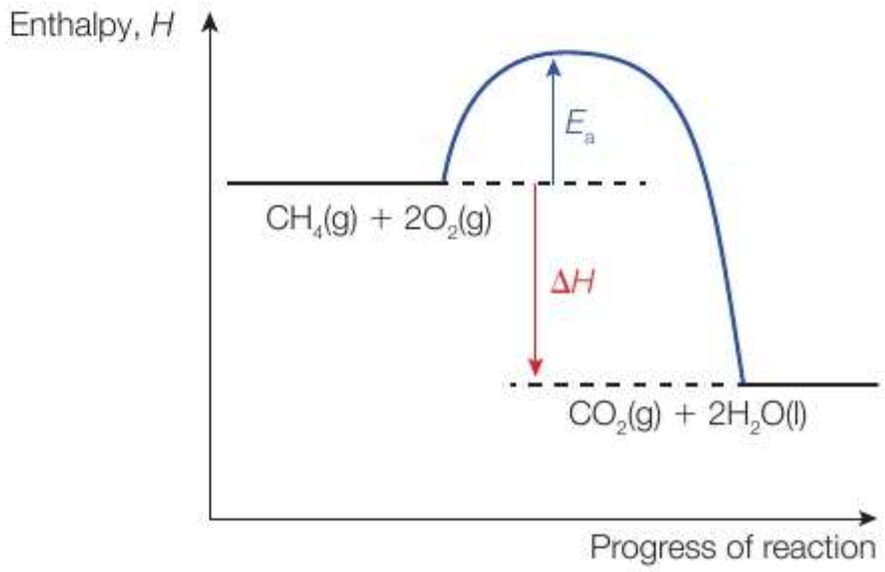
LEARNING TIP

The Maxwell–Boltzmann diagram showing the effect of a catalyst consists of *one* curve and *two* activation energy lines. Do not confuse this with the diagram used to show the effect of temperature.

REACTION PROFILE DIAGRAMS

A reaction profile diagram is an extension of an enthalpy level diagram (see **Topic 6B**). In addition to showing the relative enthalpy levels of reactants and products, it includes the activation energy for the reaction.

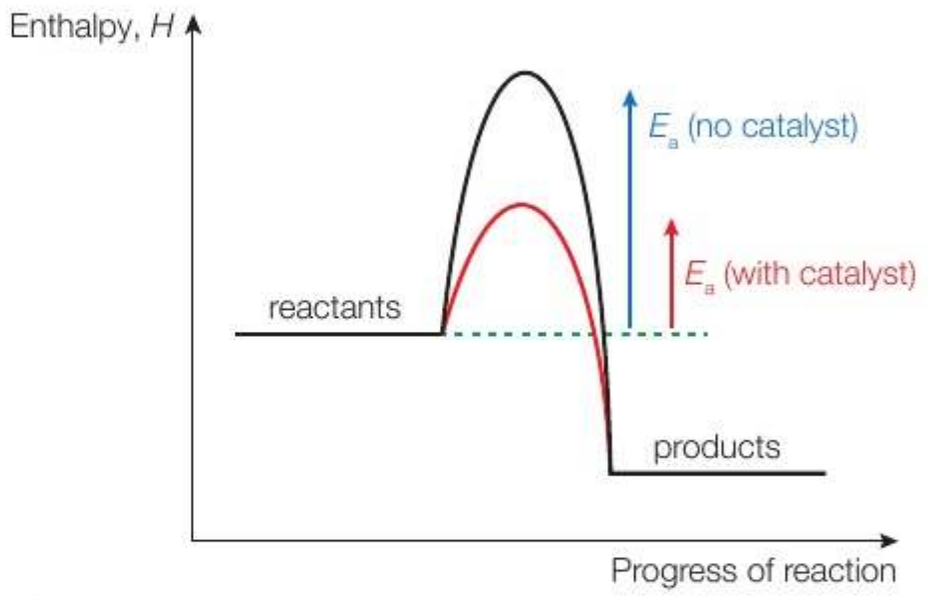
A typical reaction profile, not to scale, for the combustion of methane is shown in **fig B**.



▲ **fig B** A reaction profile for the combustion of methane.

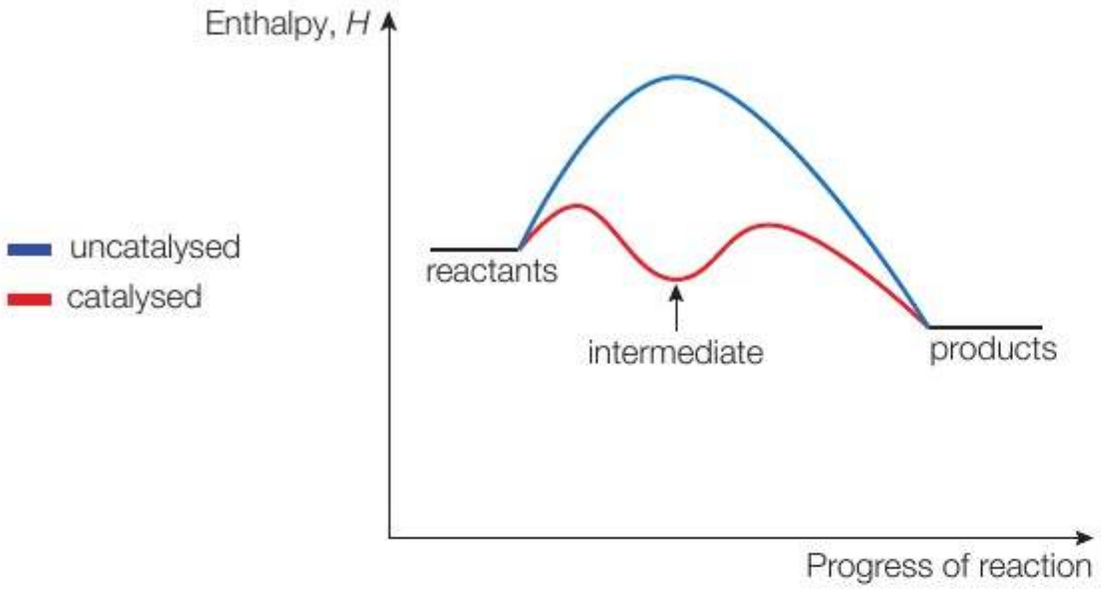
For an endothermic reaction, the enthalpy level of the products is above the enthalpy level of the reactants, but otherwise the profile is the same.

Fig C shows the simplified reaction profiles for an uncatalysed reaction and a catalysed reaction.



▲ **fig C** Simplified reaction profiles for an uncatalysed and a catalysed reaction.

If the catalysed reaction involves the formation of an intermediate, then the reaction profile is more complicated, as shown by the example in **fig D**.



▲ **fig D** Enthalpy profile diagram for a catalysed reaction involving the formation of an intermediate.

An example of a reaction that falls into this category is the reaction between peroxydisulfate ions and iodide ions, catalysed by Fe^{2+} ions.

The equation for the uncatalysed reaction is

$$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

The catalysed reaction occurs in two steps:

Step 1: $S_2O_8^{2-}(aq) + 2Fe^{2+}(aq) \rightarrow 2SO_4^{2-}(aq) + 2Fe^{3+}(aq)$

Step 2: $2Fe^{3+}(aq) + 2I^-(aq) \rightarrow 2Fe^{2+}(aq) + I_2(aq)$

The species present at the intermediate stage are $SO_4^{2-}(aq)$ and $Fe^{3+}(aq)$

N.B. The reaction can also be catalysed by Fe^{3+} ions. In this case the two steps will occur in reverse order.

CATALYSTS IN INDUSTRY

The first recorded use of a catalyst in industry was in 1746, when John Roebuck developed the lead chamber process for manufacturing sulfuric acid. Since then, catalysts have been increasingly used in the chemical industry. The development of catalysts has been mostly for economic reasons. However, more recently the development of catalysts has been for political and environmental reasons as well.

EXAM HINT

Longer exam questions may ask you to compare different industrial processes. Remember to include ideas about yield and atom economy as well as issues of energy and broader environmental issues.

The two major economic advantages of the use of catalysts are:

- they increase the rate of a chemical reaction, meaning that more of the desired product can be made in a given time period
- reactions can take place at lower temperatures, resulting in a decrease in the energy costs to the manufacturer.

Most catalysts used in industry are **heterogeneous catalysts**. A heterogeneous catalyst is one that is in a different **phase** to that of the reactants. Solids are commonly used as heterogeneous catalysts for reactions involving gases. The most well-known examples are the use of iron in the Haber process and of vanadium(V) oxide in the Contact process (see **Topic 17 (Book 2: IAL)**).

A solid catalyst provides a surface on which the gas molecules can adsorb and then react. The product molecules then desorb from the surface and more reactant molecules take their place. This process is described in more detail in **Topic 11 (Book 2: IAL)**.

DID YOU KNOW?

Environmental catalysts have also been developed to remove toxic products from industrial waste materials. Do some research to identify these catalysts.

CATALYSTS AND ATOM ECONOMY IN INDUSTRY

In the manufacture of chemicals, waste is produced when products are formed that are of little or no commercial value.

Using catalysts can lead to reactions being developed that make the desired product with little or no co-product. They may also reduce side reactions producing other unwanted products, making product separation easier.

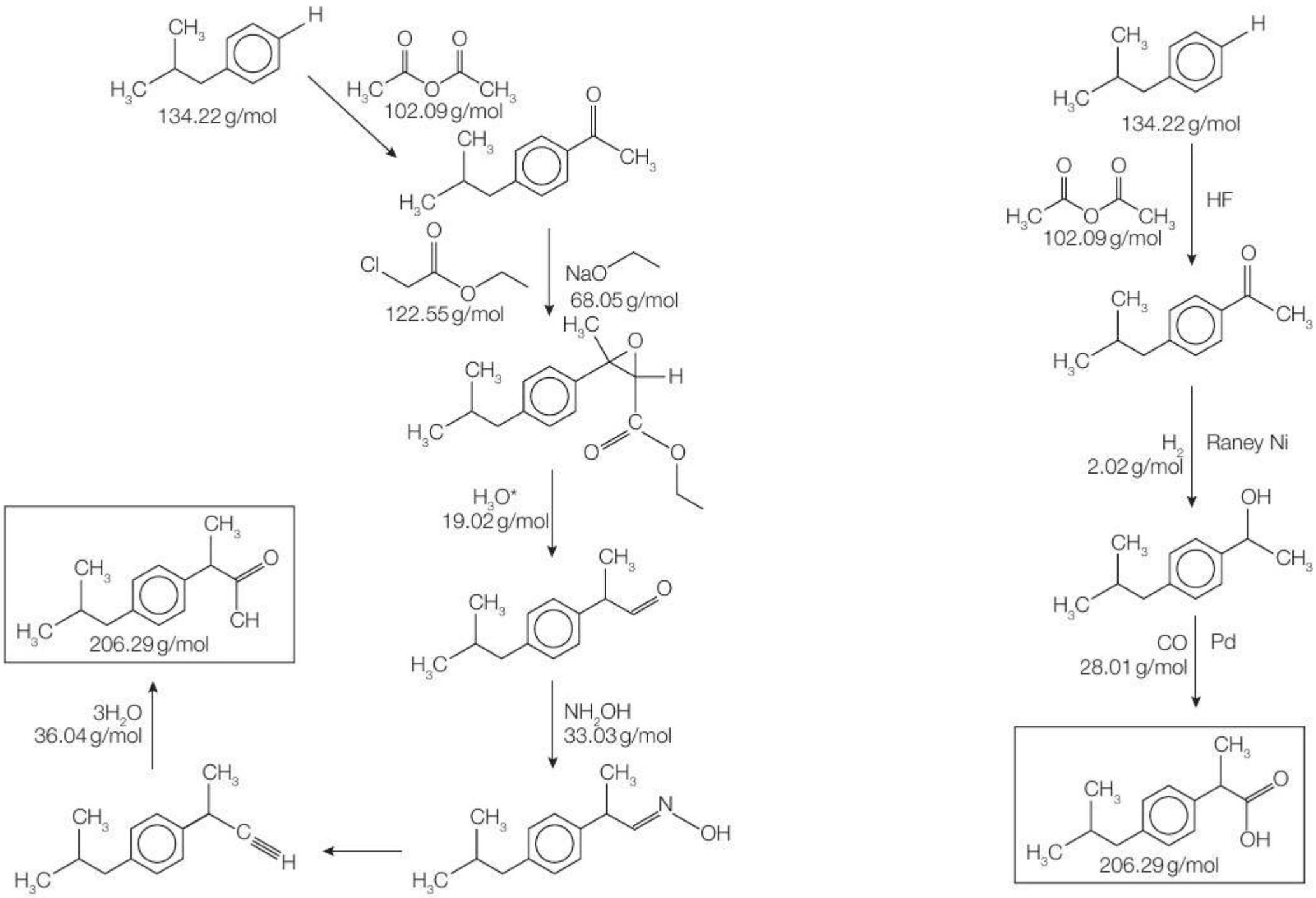
A good example is the manufacture of phenol. The development of zeolite catalysts has meant that benzene can be oxidised to phenol using dinitrogen monoxide (N_2O), with nitrogen as the only other product. The alternative manufacture from benzene via cumene has a much lower atom economy.

The manufacture of ibuprofen is another example. The original method for making it used six steps with aluminium chloride as a catalyst, and several other reagents. Although behaving as a catalyst, the aluminium chloride cannot be recovered.

A newer method uses two catalysts, hydrogen fluoride and Raney nickel (an alloy of nickel and aluminium). These can be recovered and reused many times. The effect of using better catalysts has been to improve the atom economy of the process and to reduce the number of steps needed from six to three. You can see this in **fig E**, but you don't need to know these steps.

Some other considerations are:

- With fewer steps the energy costs are likely to be less.
- The percentage yield for each step may be higher (and fewer steps).
- There may be less product lost in separation after each step or separation may not be required until the end.
- The whole process may be quicker so that you get more product in a given time.



▲ **fig E** Reaction pathways for two methods of manufacturing ibuprofen.

CHECKPOINT

1. An aqueous solution of hydrogen peroxide decomposes very slowly at room temperature into water and oxygen. The decomposition is catalysed by solid manganese(IV) oxide.

$$H_2O_2(aq) \rightarrow H_2O(l) + \tfrac{1}{2}O_2(g) \qquad \Delta_r H^{\ominus} = -196\ kJ\ mol^{-1}$$

(a) Using the same axes, draw labelled reaction profiles for the reaction both with and without manganese(IV) oxide.

(b) Explain the change in rate of reaction that occurs when manganese(IV) oxide is added to the hydrogen peroxide solution.

SUBJECT VOCABULARY

reaction pathway the reaction, or series of reactions, that the reactants undergo in order to change into the products

catalyst a substance that increases the rate of a chemical reaction but is chemically unchanged at the end of the reaction

heterogeneous catalyst a catalyst that is in a different phase to that of the reactants

phase a physically distinct form of matter, such as a solid, liquid, gas or plasma; a phase of matter is characterised by having relatively uniform chemical and physical properties